

```
1  library ieee;
2  use ieee.std_logic_1164.all;
3
4  entity sixteen_to_eightEncoder is
5      port(input3, input2, input1, input0: in std_logic_vector(3 downto 0);
6           s:      in std_logic_vector(3 downto 0);
7           en:      in std_logic;
8           output16_8En1, output16_8En0: out std_logic_vector(3 downto 0));
9  end;
10
11 architecture synth of sixteen_to_eightEncoder is
12 begin
13     process (s(3), s(2)) begin
14         if (s(3) = '1' and s(2) = '1' and en = '1') then
15             output16_8En1(3 downto 0) <= input3(3 downto 0);
16         elsif (s(3) = '1' and s(2) = '0' and en = '1') then
17             output16_8En1(3 downto 0) <= input2(3 downto 0);
18         elsif (s(3) = '0' and s(2) = '1' and en = '1') then
19             output16_8En1(3 downto 0) <= input1(3 downto 0);
20         elsif (s(3) = '0' and s(2) = '0' and en = '1') then
21             output16_8En1(3 downto 0) <= input0(3 downto 0);
22         end if;
23     end process;
24
25     process (s(1), s(0)) begin
26         if (s(1) = '1' and s(0) = '1' and en = '1') then
27             output16_8En0(3 downto 0) <= input3(3 downto 0);
28         elsif (s(1) = '1' and s(0) = '0' and en = '1') then
29             output16_8En0(3 downto 0) <= input2(3 downto 0);
30         elsif (s(1) = '0' and s(0) = '1' and en = '1') then
31             output16_8En0(3 downto 0) <= input1(3 downto 0);
32         elsif (s(1) = '0' and s(0) = '0' and en = '1') then
33             output16_8En0(3 downto 0) <= input0(3 downto 0);
34         end if;
35     end process;
36 end;
```